Yupeng Su

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Research Interests

Efficient Pre-training / Post-training Optimization for Large Language Models: focusing on quantization, sparsification, and low-rank decomposition techniques to improve training and inference efficiency.

High-Performance Computing Infrastructure on GPU / FPGA Architectures: including compiler optimization and GPU/FPGA kernel adaptation for high-performance, energy-efficient inference and practical edge deployment.

Education

PhD **University of California, Santa Barbara**, Computer Science — CA, USA, Sept 2025 – present
- Advisor: Prof. Zheng Zhang (ECE, UCSB).

BE **Southern University of Science and Technology**, Microelectronics — China, Sept 2021 – July 2025
- Advisor: Prof. Hao Yu (SME, SUSTech).
- CGA/GPA: 3.93/4.0, with 178 units completed.
- Awards: Nominee of Top Ten Graduates of SUSTech; Top Ten Graduates of the College of Engineering, SUSTech.
- Dissertation: *Enhanced Mix-Precision Post-Training Quantization for Large Language Models: Block Awareness and Outlier Identification.*

Publications

APTQ: Attention-aware post-training mixed-precision quantization for large language models — June 2024
Ziyi Guan, Hantao Huang, **Yupeng Su**, Hong Huang, Ngai Wong, Hao Yu
10.1145/3649329.3658498 (IEEE/ACM DAC, 2024)

LLM-Barber: Block-Aware Rebuilder for Sparsity Mask in One-Shot for Large Language Models — Nov 2025
Yupeng Su, Ziyi Guan, Xiaoqun Liu, Tianlai Jin, Dongkuan Wu, Chenfei Chen, Graziano Chesi, Ngai Wong, Hao Yu
10.1109/ICCAD66269.2025.11240704 (IEEE/ACM ICCAD, 2025)

EdgeLLM: A Highly Efficient CPU-FPGA Heterogeneous Edge Accelerator for Large Language Models — May 2025
Mingqiang Huang, Ao Shen, Kai Li, Haoxiang Peng, Boyu Li, **Yupeng Su**, Hao Yu
10.1109/TCSI.2025.3546256 (IEEE TCAS I: Regular Papers, 2025)

Quantization Meets Reasoning: Exploring Low-Bit Quantization Degradation for Mathematical Reasoning — Jan 2025
Zhen Li*, **Yupeng Su***, Runming Yang, Congkai Xie, Zheng Wang, Zhongwei Xie, Ngai Wong, Hongxia Yang
arxiv.org/abs/2501.03035 (Under Review (*Equal Contribution))

PTQTP: Post-Training Quantization to Trit-Planes for Large Language Models — Sept 2025
He Xiao, Runming Yang, Qingyao Yang, Wendong Xu, Zhen Li, **Yupeng Su**, Zhengwu Liu, Hongxia Yang, Ngai Wong
arxiv.org/abs/2509.16989 (Under Review)

Experience

Prof. Hongxia Yang's Lab @ PolyU, Student Research Assistant — Hong Kong, China
Nov 2024 – May 2025
7 months

- Investigated the effects of compression on reasoning and explored data-driven strategies for restoration.

Next Gen AI (NGai) Lab @ HKU, Student Research Assistant — Hong Kong, China
Aug 2024 – Feb 2025
7 months

- Contributed to lightweight networks for efficient inference under specified computing architectures.

High Performance Integrated Circuit Design Lab @ SUSTech, Summer Intern — China
June 2024 – Sept 2024
4 months

- Implemented end-to-end pipelines for compressing, optimizing, and deploying models for efficient devices.
- Published **EdgeLLM** based on the project and its results.